

# **Vaccine & Vaccination Protocols**



Submitted to:  
Dr. Tanveer Ahmed



Vaccine  
or  
Vaccination?

# Vaccine

**“A vaccine is a biological preparation that improves immunity to a particular disease. It contains certain agents that not only resemble a disease-causing microorganism but it also stimulates the body's immune system to recognize the foreign agents.”**

- Vaccine is made by weakening or killing the germs of a disease in the laboratory.
- These weakened germs, when they are injected into the animal, but do not produce that particular disease.
- Their presence in its body naturally starts boosting their immune system.
- If the same disease attacks the animal, it already has a strong defense system to fight against that disease.

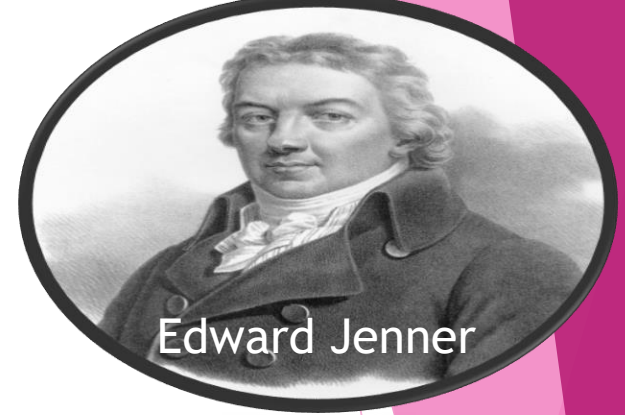


# Vaccination

- ▶ **“Vaccination is the administration of antigen material (a vaccine) to stimulate the immune system of an individual to develop adaptive immunity to a disease.”**
- ▶ Thus vaccination protects the animal against that disease.
- ▶ Due to vaccination the eradication of small pox and rinderpest and control of diseases such as FMD, rabies, influenza etc has been possible.



# History



- Vaccination is derived from Latin word “vacca” meaning from cows.
- Edward Jenner discovered vaccine first time in 1796 against smallpox by use of cowpox material to create immunity to smallpox.
- In 1879 Louis Pasteur developed a vaccine against chicken cholera.
- In 1881 anthrax vaccine was discovered by Pasteur.
- In 1885 Pasteur developed a vaccine against rabies based on live attenuated virus.

# History

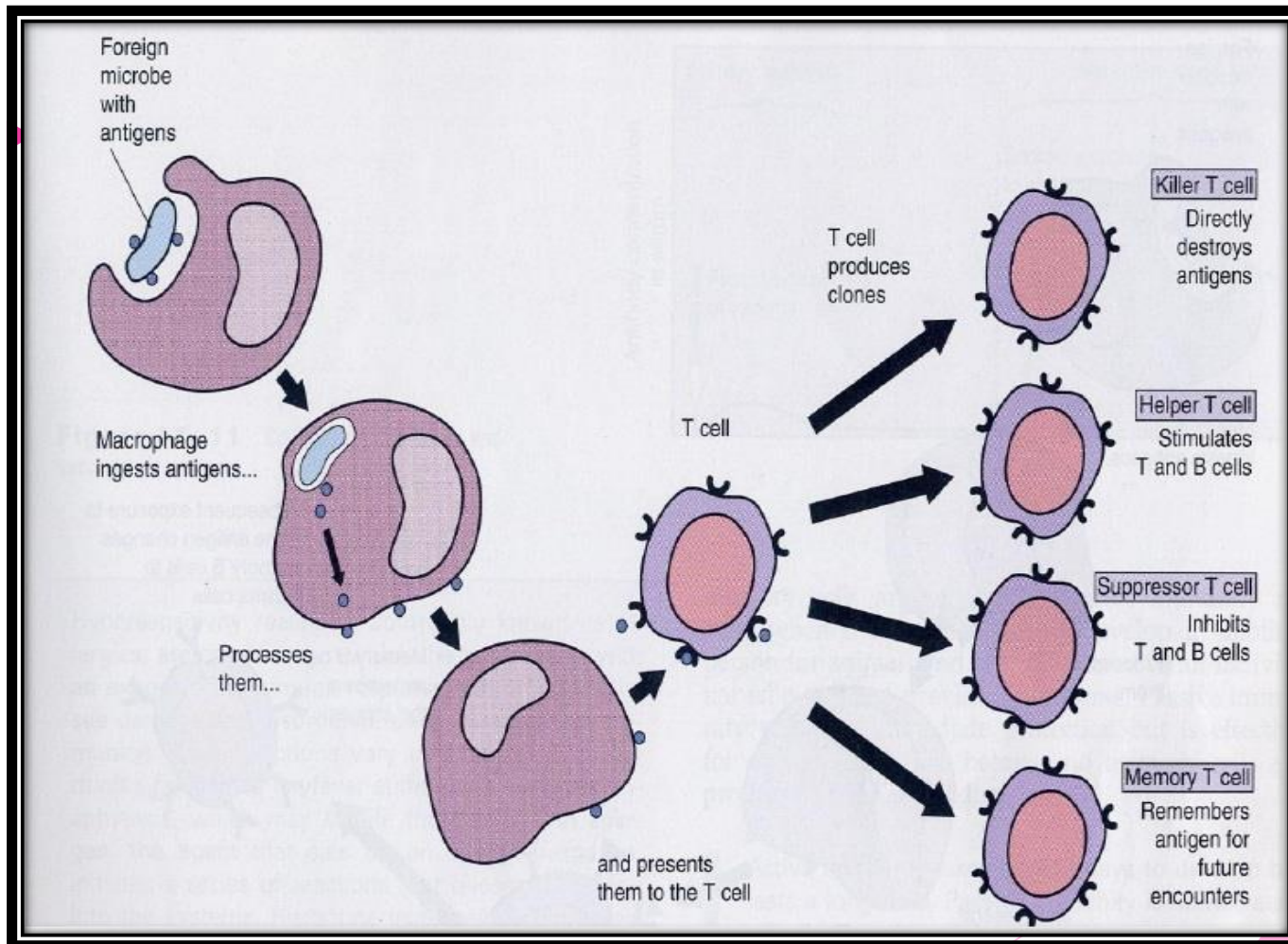
- ▶ In 1923, Alexander Glenny perfected a method to inactivate tetanus toxin with formaldehyde. The same method was used to develop a vaccine against diphtheria in 1926.
- ▶ The past two decades have seen the application of molecular genetics and its increased insights into immunology, microbiology and genomics applied to vaccinology.
- ▶ Current successes include the development of recombinant hepatitis B vaccines, the less reactogenic acellular pertussis vaccine, and new techniques for seasonal influenza vaccine manufacture.

# **Principle of Vaccination**

**“The primary goal in vaccination is to provide protective immunity by inducing memory responses to an infectious organism using a non toxic antigenic preparation.”**

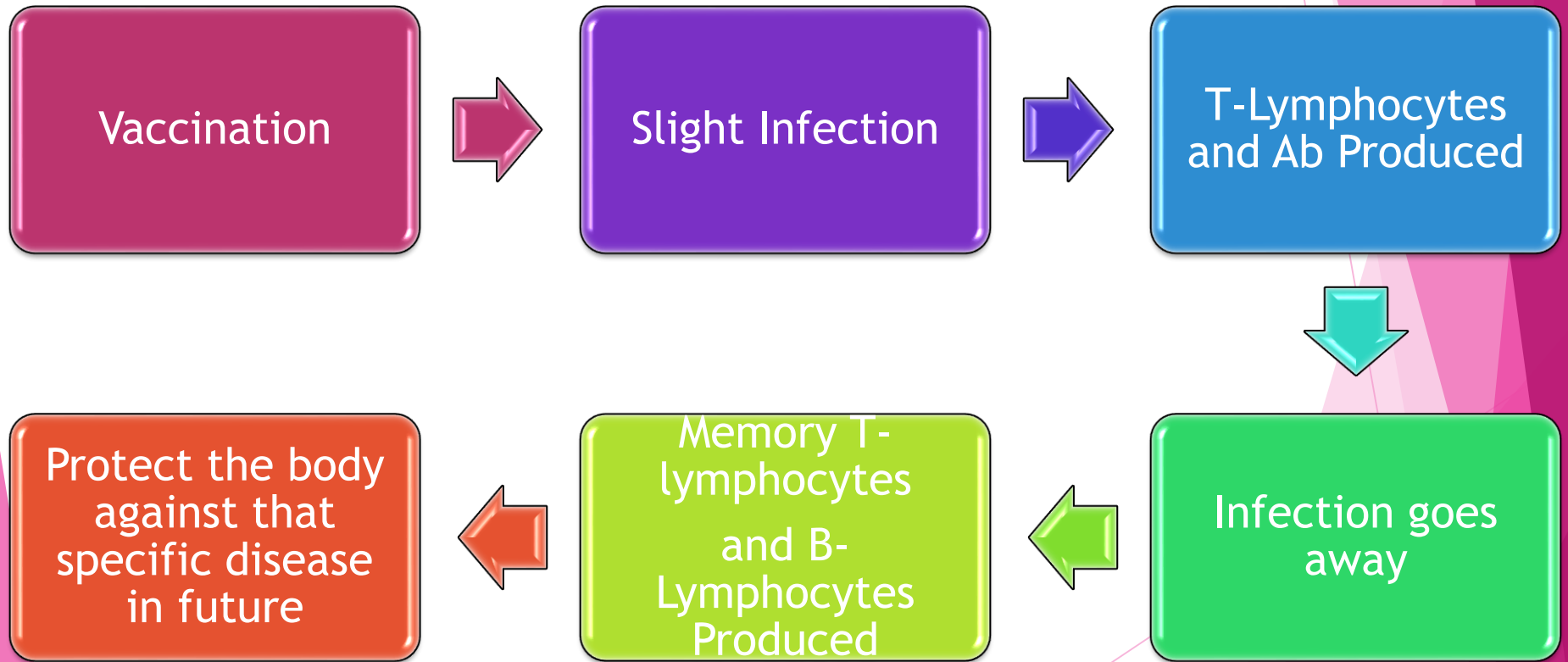


# Mechanism of Vaccine





# Vaccine Working



# Immunity

Active  
Immunity

- Natural
- Acquired

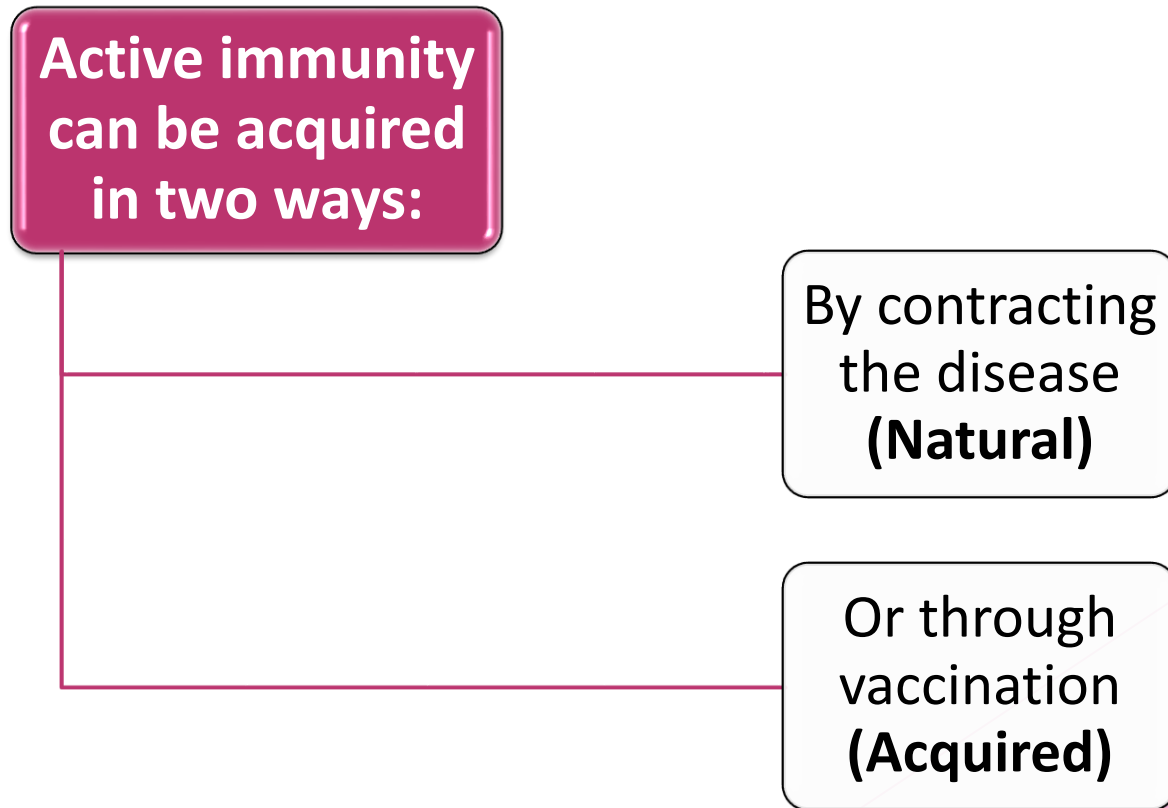
Passive  
Immunity

- Natural
- Acquired

# Active Immunity

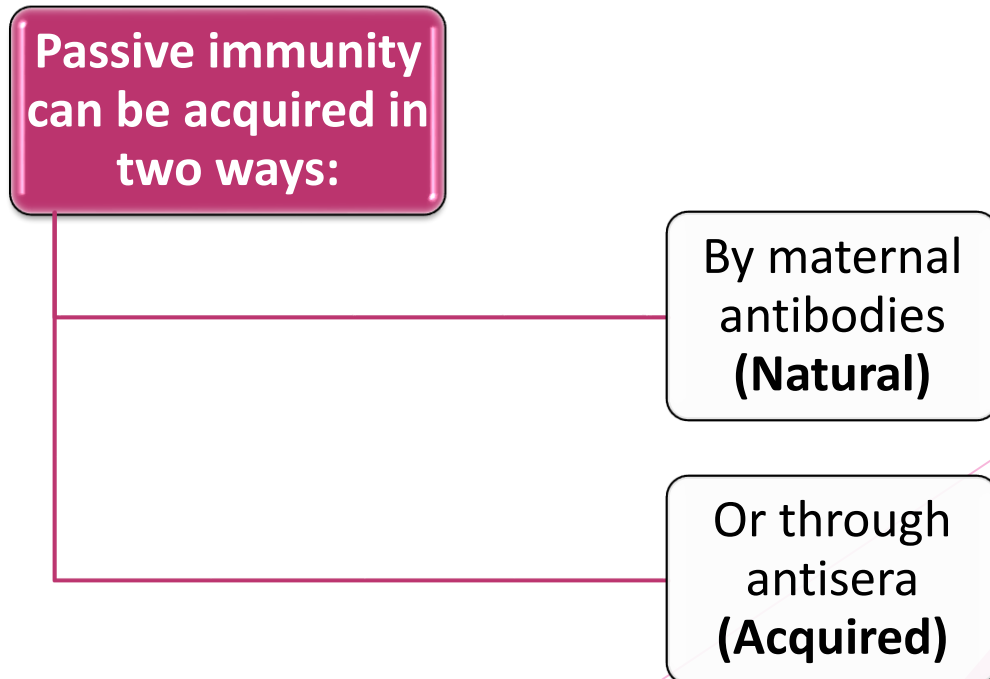
**Active immunity:** The production of antibodies against a specific disease by the immune system.

Active immunity is usually permanent, meaning an individual is protected from the disease for the duration of their lives.



# Passive Immunity

- ▶ **Passive immunity:** Protection against disease through antibodies produced by another human being or animal.
- ▶ Passive immunity is effective, but protection is generally limited and diminishes over time (usually a few weeks or months).
- ▶ For example, maternal antibodies are passed to the infant prior to birth. These antibodies temporarily protect the baby for the first 4-6 months of life.



# Vaccine Production

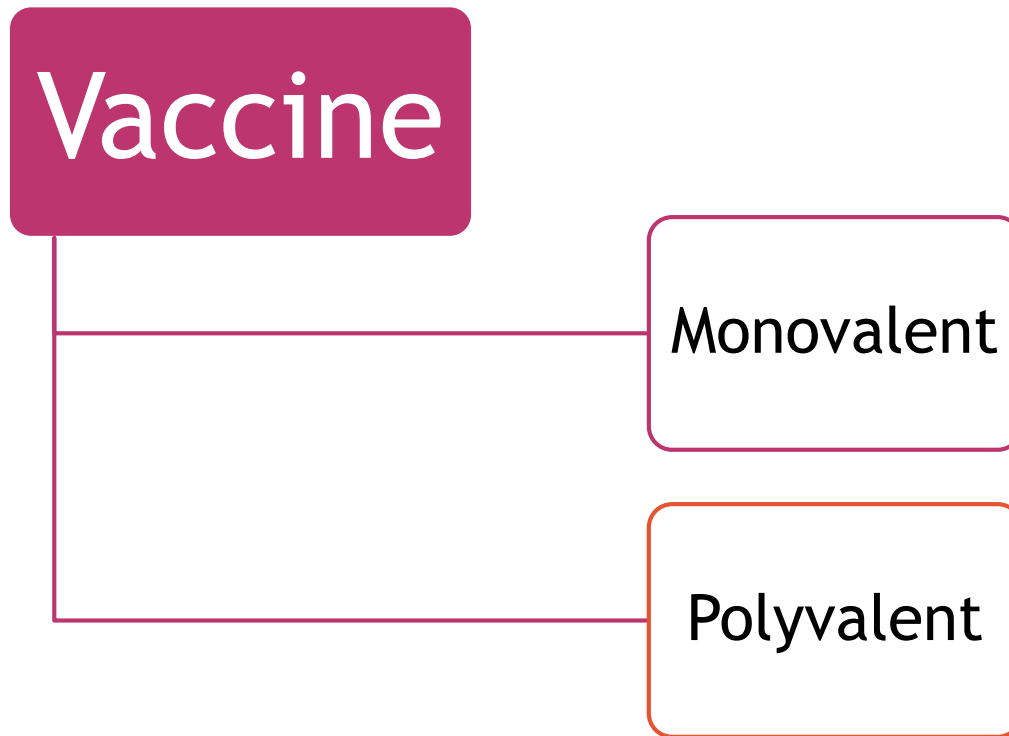
Vaccines are produced mainly in three forms:

**Liquid Vaccine** – it is in fluid form ready to use.

**Freeze Dried Vaccine** – the vaccine is stored as one pack of freeze dried material and one pack of diluent, often a sterile saline solution. These have to be combined before use.

**Dust Vaccine** – where the vaccine is prepared for administration in the dry form.

# Valence Vaccines



# Valence Vaccines

Vaccines may be monovalent (also called univalent) or multivalent (also called polyvalent).

## Monovalent Vaccine:

A monovalent vaccine is designed to immunize against a single antigen or single microorganism.

A monovalent vaccine contains a single strain of a single antigen (**e.g. Measles vaccine**)

## Polyvalent Vaccine:

A multivalent or polyvalent vaccine is designed to immunize against two or more strains of the same microorganism, or against two or more microorganism

Polyvalent vaccine contains two or more strains/serotypes of the same antigen (**e.g. OPV**).



# Combination vaccines

- ❑ Some of the antigens above can be combined in a single injection.
- ❑ It can prevent different diseases or that protect against multiple strains of infectious agents causing the same disease.
- ❑ Combination vaccine DPT combining diphtheria, pertussis and tetanus antigens.
- ❑ Combination vaccines can be useful to overcome logistic constraints of multiple injections, and accommodate for a children's fear of needles and pain.

# Vaccine Types

1. Live, attenuated vaccines
2. Inactivated vaccines
3. Subunit vaccines
4. Toxoid vaccines
5. Conjugate vaccines
6. DNA vaccines
7. Recombinant vector vaccines



# Live, Attenuated Vaccines

**“Live, attenuated vaccines contain a version of the living microbe that has been weakened in the lab so it can’t cause disease.”**

**Example:** Vaccines against measles, mumps, and chickenpox

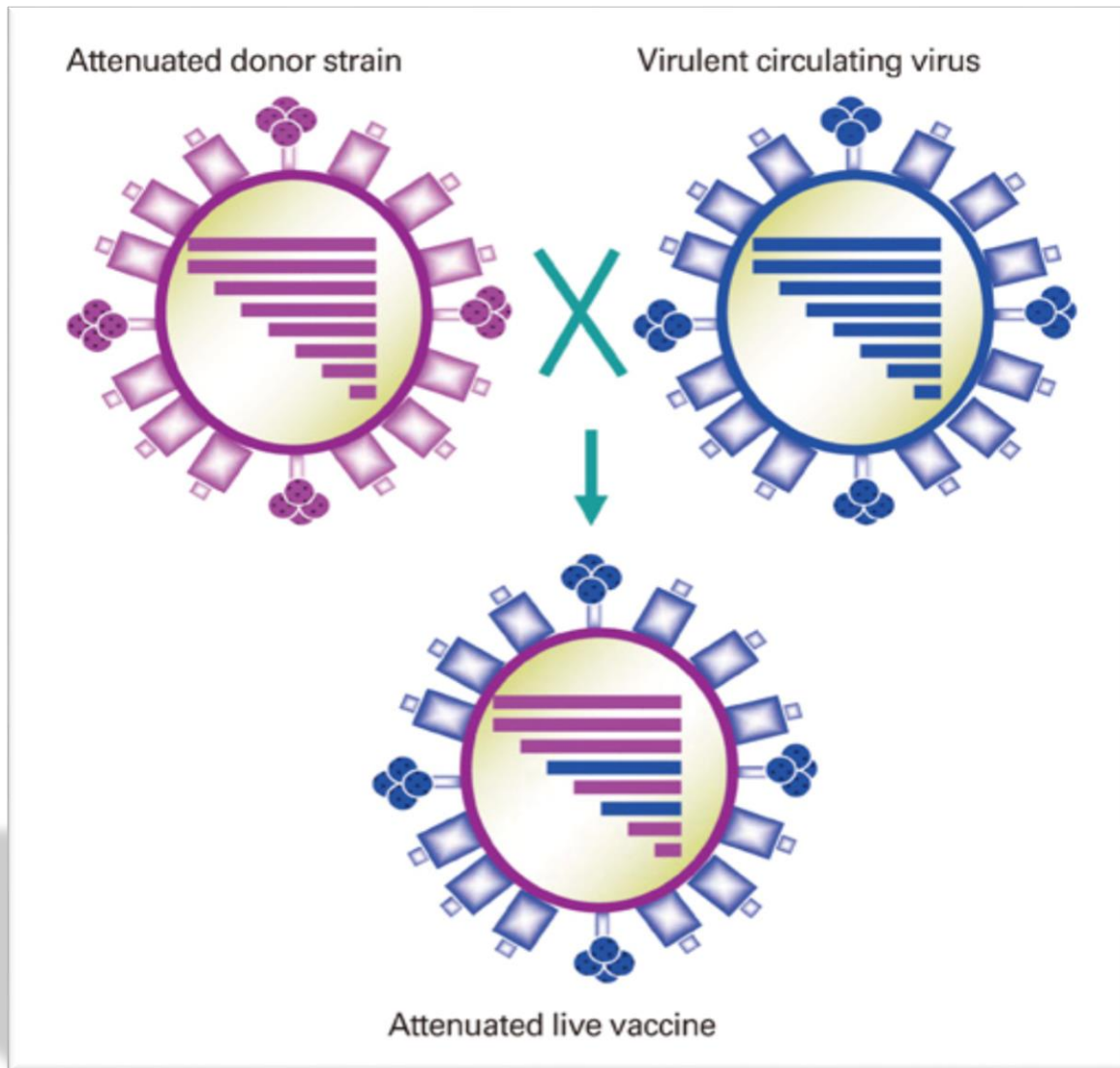
## Advantages:

- ▶ Because a live, attenuated vaccine is the closest thing to a natural infection, these vaccines are good for the immune system.
- ▶ They only require one to two doses for life-long immunity. (Fewer boosters)
- ▶ These vaccines are relatively easy to create for certain viruses

## Disadvantages:

- ❑ Ability to gain control of their natural environment again and cause disease.
- ❑ These vaccines can not be administered to people with weakened immune systems.
- ❑ Live, attenuated vaccines usually must be refrigerated and protected from light.

# Live, Attenuated Vaccines



# Killed or Inactivated Vaccines

**“Inactivated vaccines is obtained by killing the disease-causing microbe with chemicals, heat, or radiation.”**

**Example:** Vaccines against influenza, polio, hepatitis A, and rabies.

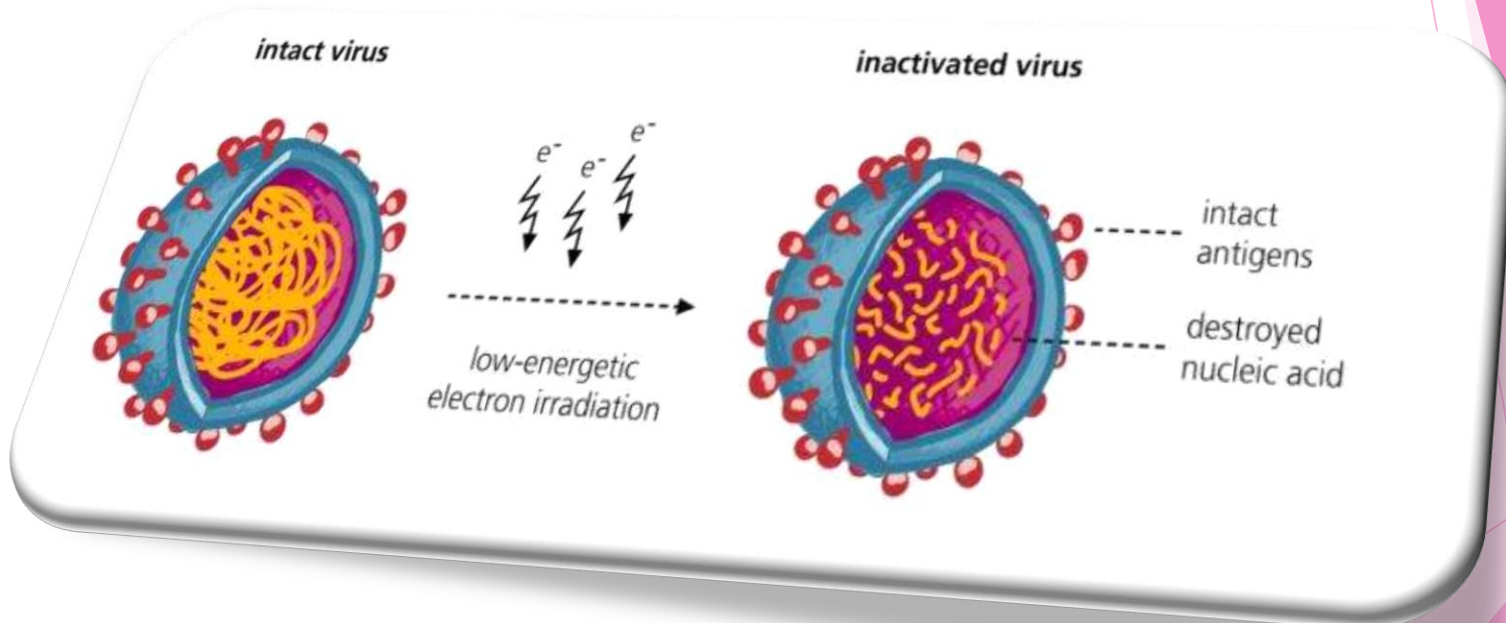
## Advantages:

- ▶ Vaccines are more stable and safer than live vaccines.
- ▶ No chance of disease causation.
- ▶ Can be given to people with weakened immune systems

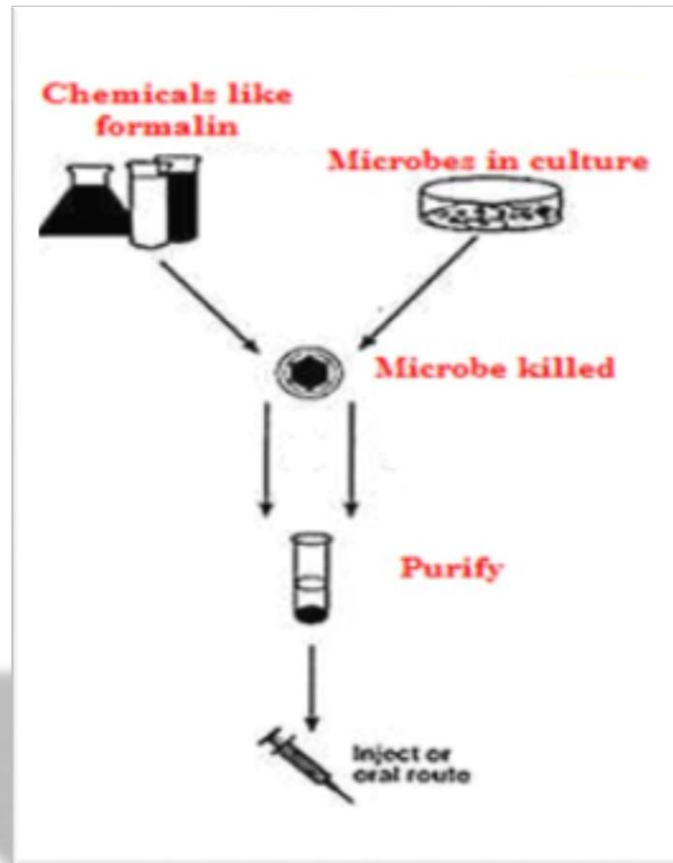
## Disadvantages:

- ❑ Stimulate a much weaker immune response than live vaccines.
- ❑ Booster shots are sometimes needed to keep immunity to the disease.

# Killed or Inactivated Vaccines



# Killed or Inactivated Vaccines





# Live Attenuated VS Killed Vaccine

| Living vaccines                                                                                                                                                                                                             | Inactivated vaccines                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>1-Few inoculating doses required.</li><li>2-Adjuvants unnecessary</li><li>3-Less chance of hypersensitivity.</li><li>4-Induction of interferon.</li><li>5-Relatively cheap.</li></ul> | <ul style="list-style-type: none"><li>1-Stable on storage</li><li>2-Unlikely to cause disease through residual virulence.</li><li>3-Unlikely to contain live contaminating organisms.</li></ul> |



# Subunits Vaccines

**“Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system.”**

**Example:** Hepatitis B Vaccine, HPV Vaccine

- ▶ In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to.

## Advantages:

- ▶ Subunit vaccines can be given to people with weakened immune systems.
- ▶ These vaccines appear to give long-lived immunity
- ▶ Since only parts of the virus are used for these vaccines, the risks of reactions are very low.

## Disadvantages:

- ❑ Several doses must be given for proper life-long immunity.

# Subunits Vaccines

**Subunit vaccines** are vaccines that use only part of the disease-causing virus



**Live Hepatitis B Virus**



**Hepatitis B Subunits**

# Toxoid Vaccines

**“Toxoid vaccine is prepared for bacteria that secrete toxins, or harmful chemicals.”**

**Example:** *Crotalus atrox toxoid* is used to vaccinate dogs against rattlesnake bites.

- ❑ These vaccines are used when a bacterial toxin is the main cause of illness.
- ❑ Scientists have found that they can inactivate toxins by treating them with formalin. Such “detoxified” toxins, called toxoids, are safe for use in vaccines.

# Conjugate Vaccines

**“Conjugate vaccines use polysaccharide (sugar) molecules are taken from the outside layer of encapsulated bacteria and join the molecules to carrier proteins.”**

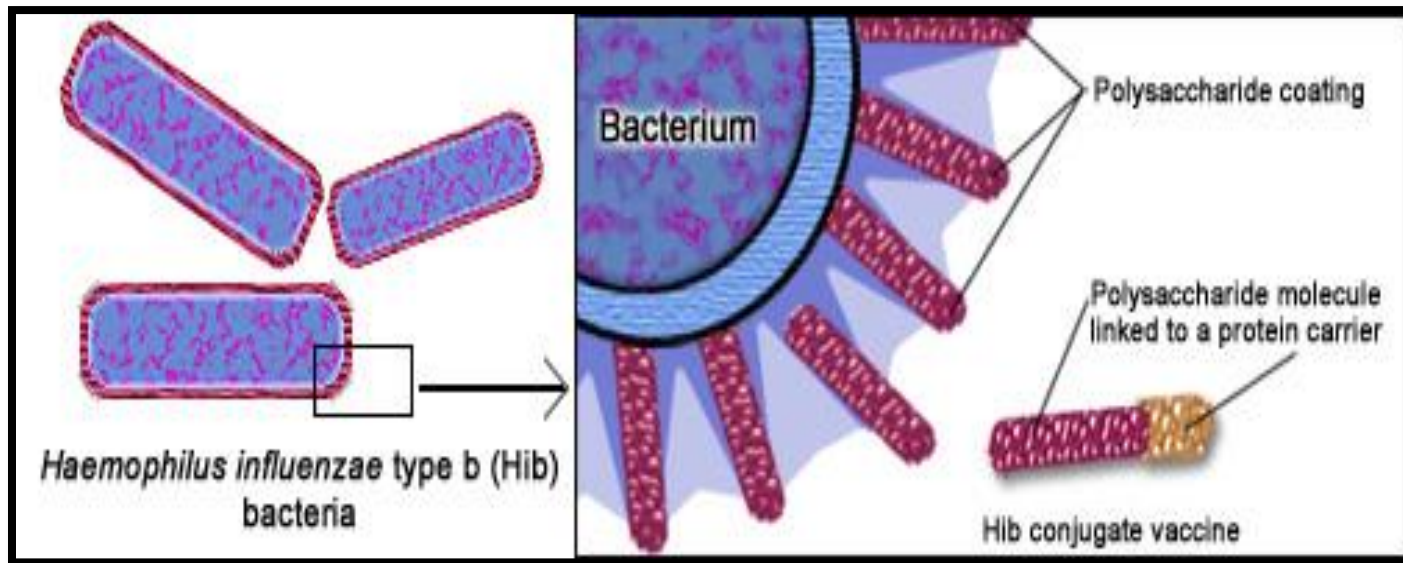
**Example :** *Haemophilus influenzae* type B vaccine.

- ❑ If a bacterium possesses an outer coating of sugar molecules called polysaccharides, as many harmful bacteria do, researchers may try making a conjugate vaccine for it.
- ❑ Polysaccharide coatings disguise a bacterium's antigens so that the immature immune systems of infants and younger children can't recognize or respond to them.

## Advantages

- ▶ Improved immune and memory response,
- ▶ Longer lasting protection
- ▶ The protection of infants and toddlers
- ▶ The creation of herd immunity

# Conjugate Vaccines



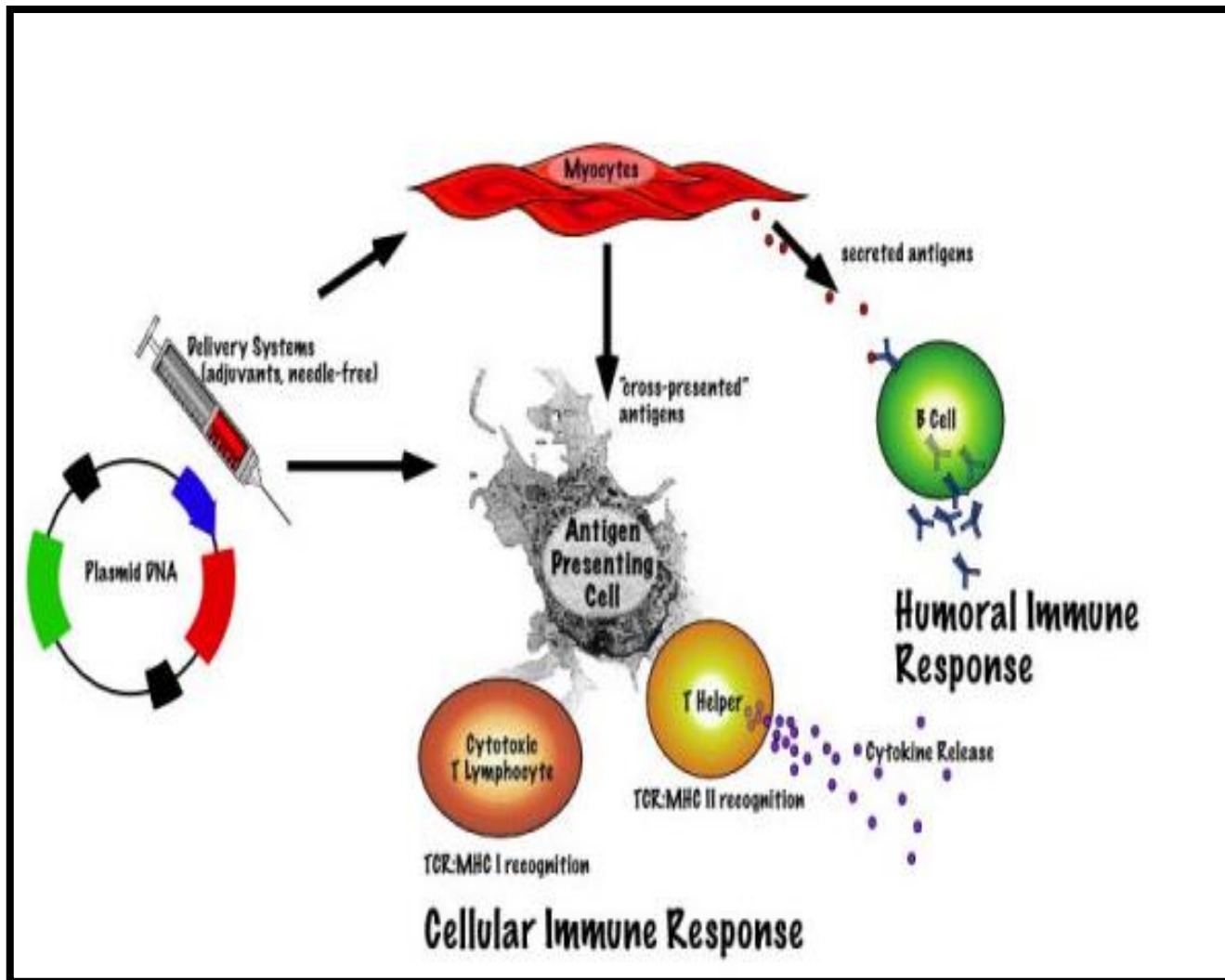
# DNA Vaccines

**“Once the genes from a microbe have been analyzed, scientists could attempt to create a DNA vaccine against it.”**

**Example:** Influenza vaccine.

- ▶ Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans.
- ▶ DNA vaccines take immunization to a new technological level.
- ▶ These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe’s genetic material.

# DNA Vaccines



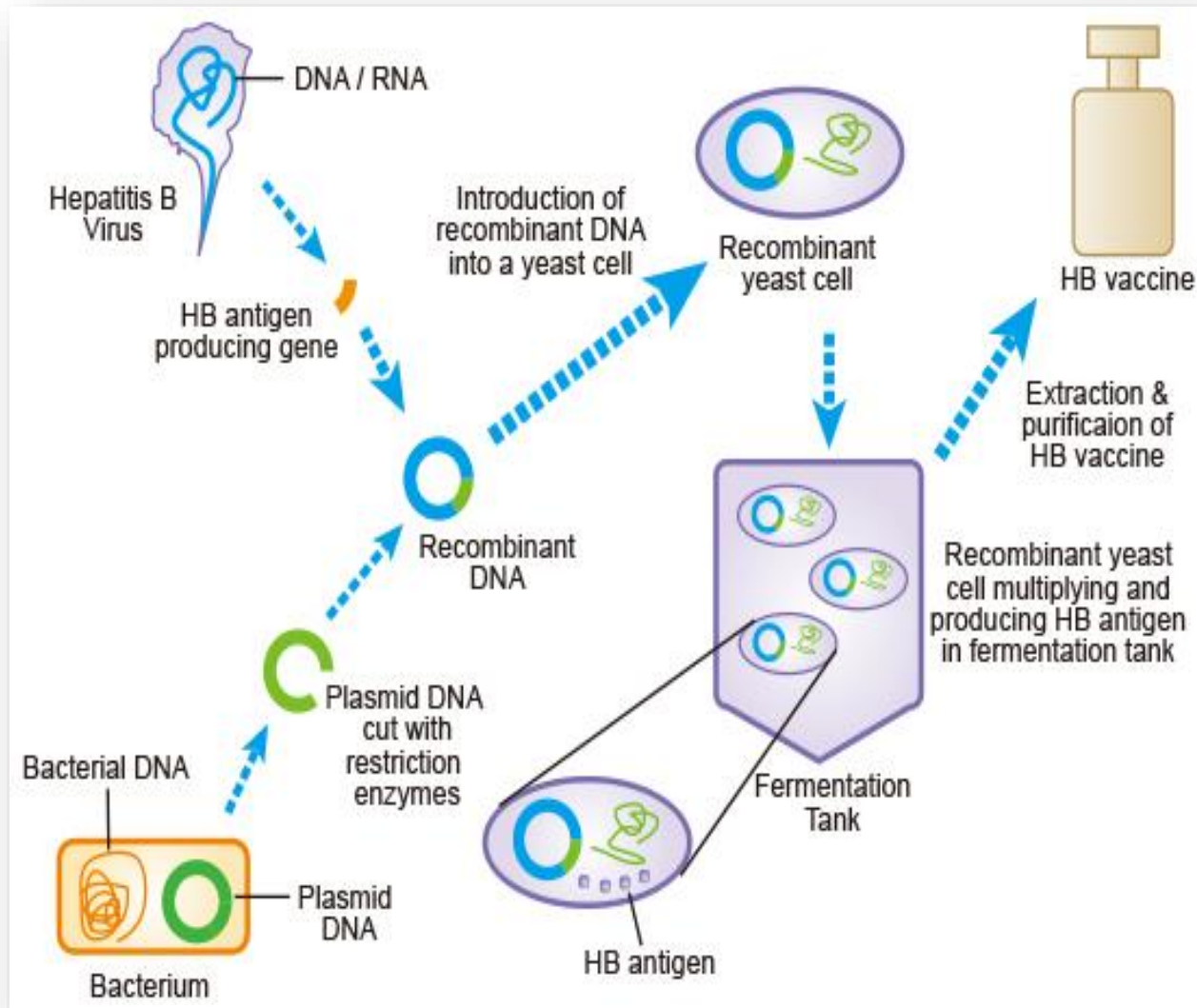
# Recombinant Vector Vaccines

The gene segment for a protein from the disease-causing organism that is known to stimulate a protective immune response (protein of interest) is inserted into the gene of another cell, such as a yeast cell. When the cell replicates it has the same shape as the protein of interest.

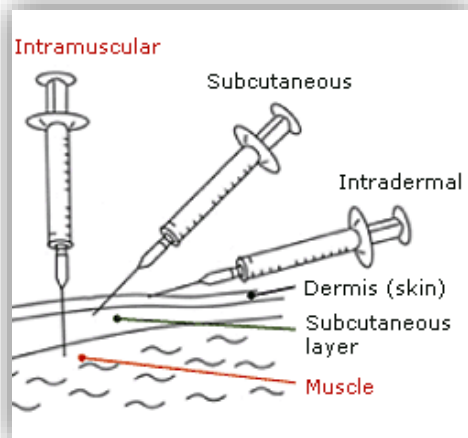
- ❑ Yeasts cells are used for the hepatitis B
- ❑ Recombinant vector vaccines are experimental vaccines similar to DNA vaccines.
- ❑ But they use an attenuated virus or bacterium to introduce microbial DNA to cells of the body.
- ❑ “Vector” refers to the virus or bacterium used as the carrier.



# Recombinant Vector Vaccines



# Routes of Vaccine Administration in Poultry



# Routes of Vaccine Administration in Poultry

## ❑ In-ovo Vaccination:

- Using the method of in-ovo vaccination, the vaccine is administered into the embryo before hatch.
- In-vivo vaccination is commonly performed between days **18-19** of incubation.
- Currently Marek's disease, Newcastle disease, infectious laryngotracheitis and infectious bursal disease vaccines are routinely administered using in ovo vaccination in various countries.

## ❑ Injections:

- Subcutaneous, intra muscular depending on the type of vaccine and the direction of the manufacturer.
- Subcutaneous, injection to administer either live or inactivated emulsion vaccines to chicks rearing stocks and breeders

# Routes of Vaccine Administration in Poultry

## ❑ INTRANASAL:

- By instillation or dipping the beak up to the nares.
- In birds nose, mouth and lachrymal duct opened in mouth. In these route please do not dip the beak for too long or short time.

## ❑ INTRAOCULAR:

- Eye dropping is most popular route of vaccination. Eye drop and intranasal routes suitable for hatchery administration and during brooding of chicks.
- In Pakistan the eye droppers are 30ml for 1000 birds, the vaccinator should not hold vaccine for too long time.
- Contaminated diluents or vaccines leads to infection of eye and blocked of lachrymal duct and drooling of water from the eye used for vaccination.
- This reaction may some time normally occur after vaccination of ILT.

# Routes of Vaccine Administration in Poultry

## ❑ IN DRINKING WATER:

- Drinking water administration, can be implemented at low cost but it is of limited effectiveness against some infections.

## ❑ POST-HATCH SPRAY VACCINATION:

- In cabinets for mass-administration of aerosol vaccines to day old chicks. This is mostly employed for ND, IB vaccination on day one in hatchery or farm.

## ❑ WING-WEB STAB:

- This method is used to administer fowl pox vaccine or live vaccines directly to each bird. Be careful that eye of the needle carries vaccine each time or not.

## ❑ AEROSOL ADMINISTRATION:

- Using a knapsack or electric sprayer to deliver vaccines to flocks as a coarse spray.

# Routes of Vaccine Administration in Poultry

## ❑ Cloacal

- This method involves the introduction of the vaccine to the mucus membranes of the cloaca with an abrasive applicator.
- The applicator is firstly inserted into the vaccine and then into the bird's cloaca and turned or twisted vigorously to cause an abrasion in the organ. The vaccine enters the body through the abrasion.
- As a rule, the technique is not used on commercial farms.

## ❑ Feather follicle

- With this method the vaccine is introduced into the feather follicles (the holes in the skin from where the feathers grow).
- The technique involves the removal of a group of adjacent feathers or fluff in young chickens, and the brushing of the vaccine into the empty follicles with a short, stiff bristled brush.

# Routes of Vaccine Administration in Animal

## ❑ Intramuscular

E.g. Hemorrhagic septicemia

## ❑ Subcutaneous

Most of the vaccines are given by subcutaneous route e.g. FMD, Rinderpest, Anthrax, Black Quarter, Parvo, Canine Distemper etc.

## ❑ Intradermal

E.g. Tuberculosis

## ❑ Intranasal

E.g. Kennel Cough



# Cattle and Buffaloes Vaccination





# Vaccination Schedule for Cattle and Buffaloes

| Disease              | Vaccine                           | Age       | Dose     | Booster         | Interval   | Season                    |
|----------------------|-----------------------------------|-----------|----------|-----------------|------------|---------------------------|
| <b>FMD</b>           | FMD vaccine                       | 6-8 weeks | 10ml s/c | 6,9 or 12 month | Annual     | Nov, Dec                  |
| <b>Rinderpest</b>    | Rinderpest tissue culture vaccine | All Ages  | 1ml s/c  | -----           | Every year | Winter                    |
| <b>Anthrax</b>       | Anthrax spore vaccine             | All Age   | 1ml s/c  | 6 Months        | Annual     | Feb/March /April/May      |
| <b>Black Quarter</b> | Polyvalent B.Q vaccine            | All Age   | 5ml s/c  | 6 Months        | Annual     | All seasons endemic areas |

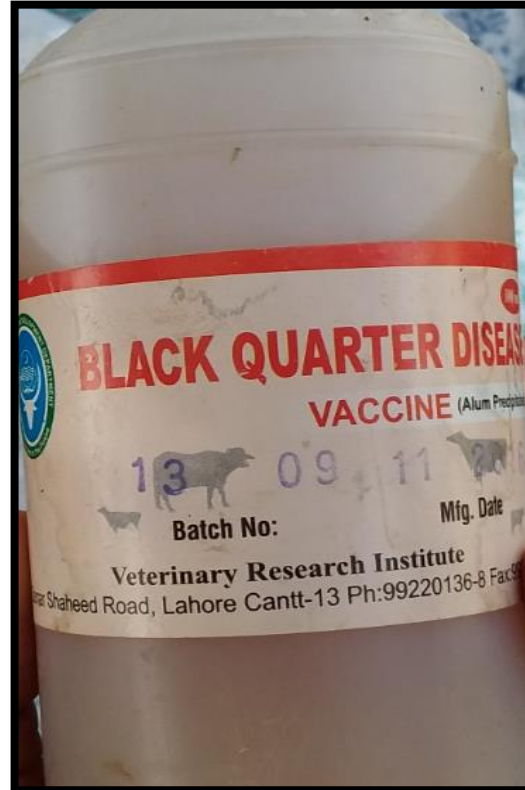
# Vaccination Schedule for Cattle and Buffaloes (Continued)

| Disease                       | Vaccine                                 | Age                   | Dose    | Booster  | Interval                       | Season                      |
|-------------------------------|-----------------------------------------|-----------------------|---------|----------|--------------------------------|-----------------------------|
| <b>Hemorrhagic septicemia</b> | Hemorrhagic Septicemia Adjuvant Vaccine | All Age               | 3ml I/m | 6 Months | Annual                         | May ,June                   |
| <b>Brucellosis</b>            | Strain 19 (live)                        | About 6 months of age | 2ml s/c | 6 Months | 3-4 calving's                  | To be done in infected herd |
| <b>Theleriosis</b>            | Attenuated, cultured vaccine            | Above 2 months of age | 3ml s/c | 6 Months | -----                          | Rainy Season                |
| <b>Tuberculosis</b>           | BCG                                     | About 6 months of age | 5ml/ID  | 6 Months | To be repeated every 2-3 years | Dec, Jan                    |

# Vaccines for Cattle and Buffaloes



**FMD Vaccine**

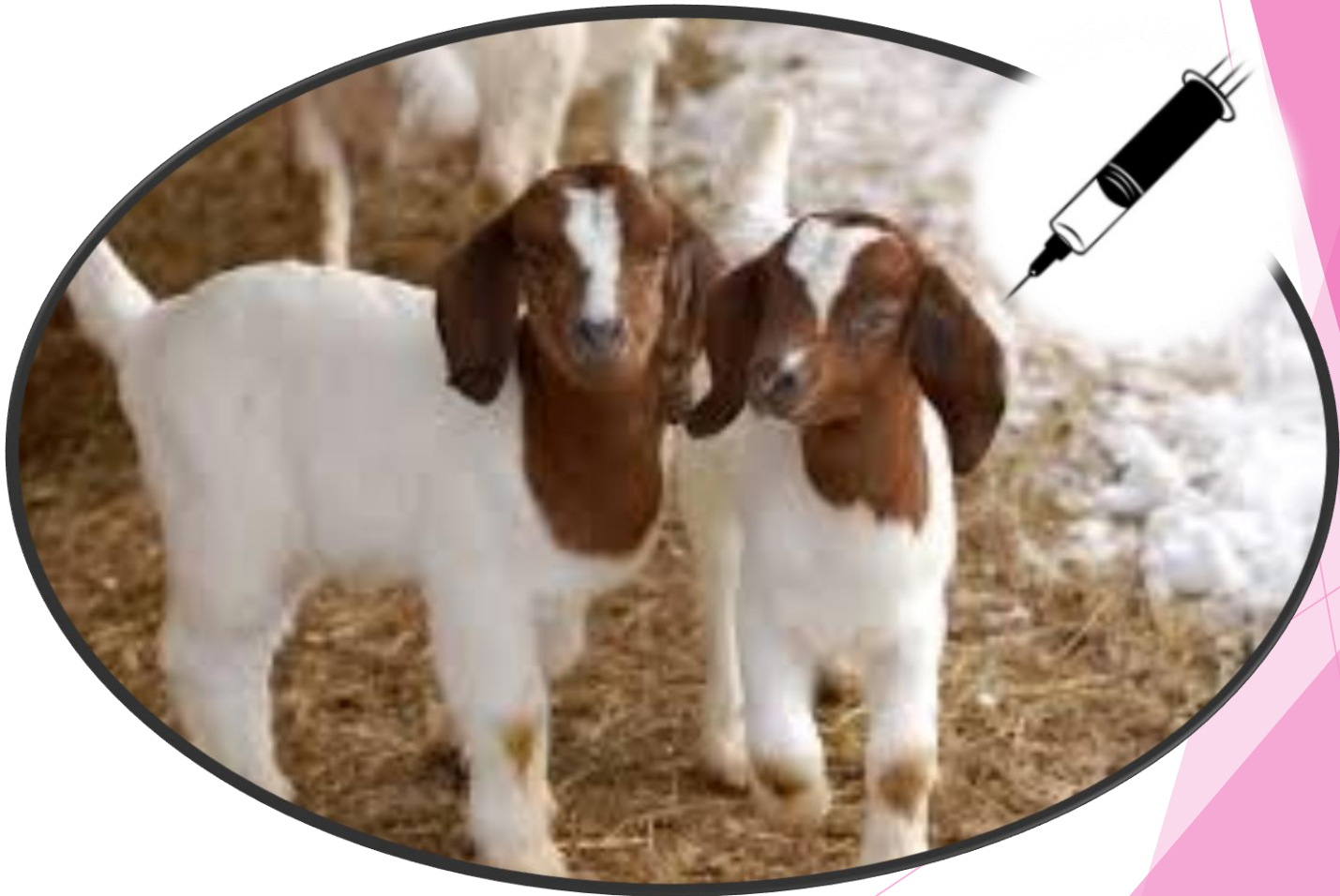


**BQ Vaccine**



**Enterotoxaemia Vaccine**

# Goat Vaccination



# Vaccination Schedule for Goats

| Months              | Vaccine                                        | Adult Goat         | Kids<br>Above 6 Months |
|---------------------|------------------------------------------------|--------------------|------------------------|
| January             | <b>Contagious<br/>pleuro-pneumonia</b>         | 0.2ml I/D          | 0.2ml I/D              |
| March               | <b>Hemorrhagic<br/>Septicemia<br/>Goat pox</b> | 5ml s/c            | 2.5ml s/c              |
| April               | <b>Enterotoxaemia</b>                          | Scratch method     | Scratch method         |
| May                 | <b>F.M.D<br/>Rinderpest</b>                    | 5ml s/c<br>5ml s/c | 2.5ml s/c<br>5ml s/c   |
| June                | <b>Black Quarter</b>                           | 1ml s/c            | 1ml s/c                |
| July                | <b>F.M.D</b>                                   | 5ml s/c            | 0.5ml s/c              |
| August<br>September | <b>Enterotoxaemia</b>                          | 5ml s/c<br>5ml s/c | 0.5ml s/c<br>2.5ml s/c |



# Sheep Vaccination



# Vaccination Schedule for Sheep

| Diseases              | vaccine                           | Age           | Dose               | Booster  | Interval | Season                          |
|-----------------------|-----------------------------------|---------------|--------------------|----------|----------|---------------------------------|
| <b>F.M.D</b>          | F.M.D Vaccine                     | Adult         | 5ml s/c            | -----    | Annual   | Preferably winter/Autumn        |
| <b>Rinderpest</b>     | Rinderpest tissue culture vaccine | -----         | 1ml s/c            | 6 Months | Annual   | Winter                          |
| <b>Black Disease</b>  | Black disease vaccine             | -----         | 2ml s/c            | -----    | Annual   | All Seasons                     |
| <b>Lamb Dysentery</b> | Lamb dysentery Vaccine            | Lamb          | 2ml s/c            | -----    | Annual   | All Seasons                     |
| <b>Black quarter</b>  | Black Quarter vaccine(polyvalent) | Lamb<br>Adult | 2ml s/c<br>3ml s/c | -----    | Annual   | All Seasons preferably May/June |

# Vaccination Schedule for Sheep (Continued)

| Diseases                          | vaccine                            | Age                    | Dose                         | Booster                                     | Interval | Season            |
|-----------------------------------|------------------------------------|------------------------|------------------------------|---------------------------------------------|----------|-------------------|
| <b>Enterotoxaemia</b>             | Enterotoxaemia vaccine             | Lamb<br>Adult          | 2.5 to 5ml<br>s/c            | 7-10 days                                   | Annual   | Lambing<br>Season |
| <b>Hemorrhagic<br/>Septicemia</b> | HS<br>Adjuvant<br>Vaccine          | Adult                  | 2ml s/c                      | -----                                       | Annual   | March/June        |
| <b>Sheep pox</b>                  | Sheep pox<br>vaccine               | Lamb<br>Sheep          | 3ml s/c<br>5ml s/c           | Repeat at 6<br>Months<br>Repeat<br>Annually | -----    | Dec/March         |
| <b>Lung worm</b>                  | Irradiated<br>lung Worm<br>vaccine | Lamb 3<br>month<br>age | 1000<br>larvae<br>first dose | 1 Month<br>later 2000<br>larvae             | -----    | -----             |



# Dog Vaccination



# Vaccination Schedule for Dog

| Viruses                                   | Vaccine Type                                                          | Age                      | Booster            | Repeat           |
|-------------------------------------------|-----------------------------------------------------------------------|--------------------------|--------------------|------------------|
| <b>Canine Distemper and Measles Virus</b> | Modified live virus                                                   | 6-8 Weeks                | 12-16 week         | Annual           |
| <b>Infectious Canine Hepatitis Virus</b>  | Modified Live Virus(CAV-1or CAV-2)<br>Or<br>inactivated(CAV-1)vaccine | 6-8 Weeks                | 12-16 week         | Annual           |
| <b>Rabies Virus</b>                       | Modified live<br>Inactivated virus                                    | 3-4 Months<br>3-4 Months | 1 year<br>3-4 Week | 3 year<br>Annual |
| <b>Canine Leptospirosis</b>               | Killed<br>Bacteria(Bacterin)                                          | 9 Weeks                  | 12-16 Week         | Annual           |
| <b>Parvovirus</b>                         | Inactivated                                                           | 6-8 Weeks                | 12 week            | Annual           |

# Available Dog Vaccines in Market

| TYPE                                                                                     | TRADE NAME      | COMPANY              |
|------------------------------------------------------------------------------------------|-----------------|----------------------|
| Monovalent (Rabies)                                                                      | <u>Rabisin</u>  | Merial               |
| Multivalent (Parvo + Distemper + Hepatitis + Leptospirosis + Adenovirus)                 | <u>Duramune</u> | Boehringer Ingelheim |
| Multivalent (Parvo + Distemper + Hepatitis + Leptospirosis + Adenovirus + Rabies)        | <u>Hexadog</u>  | Merial               |
| Multivalent (Parvo + Distemper + Hepatitis + Parainfluenza + Leptospirosis + Rabies)     | <u>Eurican</u>  | Merial               |
| Monovalent (Parvo)                                                                       | <u>Primodog</u> | Merial               |
| Multivalent (Distemper + Hepatitis + Laryngotracheitis + Parvo + Parainfluenza + Rabies) | <u>Biocan</u>   | Bioveta              |
| Multivalent (Distemper + Herpes + Parvo + Parainfluenza)                                 | <u>Canvac</u>   | Dyntec               |

# Available Dog Vaccines in Market



**Biocan DHPPi + L**



**Biocan DHPPi + Biocan LR**

**Company Bioveta**

# Available Dog Vaccines in Market



**Hexadog CHP + LR**



**Hexadog CHP + LR**

**Company Merial**



# Available Dog Vaccines in Market



**Nobivac DHPPi**



**Nobivac Rabies**



**Nobivac Lepto**



**Nobivac KC**

**Company Nobivac**

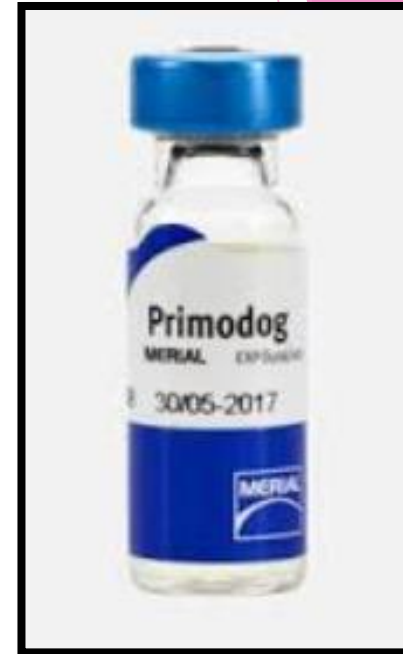
# Available Dog Vaccines in Market



**Eurican DHPPI + LR**



**Rabisin**



**Primadog**

**Company Merial**

# Cat Vaccination





# Vaccination Schedule for Cat

| Disease                                                 | Vaccine                      | Age       | Booster    | Repeat |
|---------------------------------------------------------|------------------------------|-----------|------------|--------|
| <b>Feline Panleukopaenia, Calicivirus, Herpes Virus</b> | Modified live virus Vaccine  | 8-9 Weeks | 12-16 Week | Annual |
| <b>Feline Viral Rhinotracheitis</b>                     | Modified live virus vaccine  | 8-9 Weeks | 12-16 Week | Annual |
| <b>Rabies</b>                                           | Modified live or inactivated | 12 Weeks  | -----      | Annual |

# Available Cat Vaccines in Market

| TYPE                                                                                                     | TRADE NAME           | COMPANY              |
|----------------------------------------------------------------------------------------------------------|----------------------|----------------------|
| Monovalent (Rabies)                                                                                      | <u>Rabisin</u>       | Merial               |
| Multivalent (Feline rhinotracheitis + Feline calicivirus + Feline panleukopenia + Feline leukemia virus) | <u>Fel – o – Vax</u> | Boehringer Ingelheim |
| Feline panleukopenia (feline enteritis)                                                                  | <u>Feliniffa</u>     | Merial               |
| Multivalent (Feline Panleukopenia + Feline Herpes virus + Feline Calicivirus)                            | <u>Biofel</u>        | Bioveta              |

# Available Cat Vaccines in Market



**Biofel PCH**



**Biofel PCHR**

**Company Biofel**

# Available Cat Vaccines in Market



**Fel-o-vax PCT**



**Fel-o-vax Lv-k**



**Fel-o-vax IV**

**Company Fel-O-Vax**

# Available Cat Vaccines in Market



**FELOCELL 3**



**FELOCELL 4**

**Company Zoetis**

# Available Cat Vaccines in Market



**Rabisin**

**Company Merial**



# Poultry Vaccination



# BROILER VACCINATION SCHEDULE

| AGE   | VACCINE                       | Dose/Bird<br>(ml) | ROUTE                            |
|-------|-------------------------------|-------------------|----------------------------------|
| 5-7   | ND                            | 0.025             | Eye Drop                         |
| 8-12  | Gumboro                       | 0.025             | Eye Drop                         |
| 16    | Hydro pericardium<br>Syndrome | 0.50              | Intramuscular or<br>Subcutaneous |
| 20    | Gumboro                       | 0.50              | Drinking Water                   |
| 20-22 | ND(Lasota)                    | 0.50              | Drinking Water                   |
| 28-30 | Gumboro                       | 0.50              | Drinking Water                   |



# LAYER VACCINATION SCHEDULE

| Age (Days) | Vaccine    | Dose/Bird (ml) | Route |
|------------|------------|----------------|-------|
| Day old    | Marek`s    | 0.3            | S/C   |
| 5-7        | ND+IB      | 0.025          | E/D   |
| 8-12       | Gumboro    | 0.025          | E/D   |
| 18-20      | Gumboro    | 0.025          | D/W   |
| 21-22      | ND(Lasota) | 0.025          | D/W   |
| 32-34      | Gumboro    | 0.025          | D/W   |

## LAYER VACCINATION SCHEDULE (Continued)

| Age (Days)            | Vaccine           | Dose/Bird (ml)   | Route          |
|-----------------------|-------------------|------------------|----------------|
| 38-40                 | <b>ND(Lasota)</b> | 0.025            | D/W            |
| 7 <sup>th</sup> week  | <b>Fowl pox</b>   | One dip of prong | W/W (wing web) |
| 8 <sup>th</sup> week  | <b>Coryza</b>     | 0.25             | S/C, I/M       |
| 11 <sup>th</sup> week | <b>ND+IB</b>      | 0.25             | D/W            |
| 14 <sup>th</sup> week | <b>Coryza</b>     | 0.25             | S/C, I/M       |
| 16 <sup>th</sup> week | <b>ND+IB+EDS</b>  | 0.50             | S/C, I/M       |
| After every 2 month   | <b>ND(Lasota)</b> | 0.50             | D/W            |

# BREEDER VACCINATION SCHEDULE

| Age(day)            | Vaccine         | Dose/Bird(ml) | Route   |
|---------------------|-----------------|---------------|---------|
| Just after hatching | Marek`s         | 0.3           | S/C     |
| 2 <sup>nd</sup>     | ND+IB           | 0.025         | E/D     |
| 10 <sup>th</sup>    | ND(Lasota)      | 0.025         | E/D     |
| 14 <sup>th</sup>    | Gumboro         | 0.025         | E/D     |
| 18 <sup>th</sup>    | Avian influenza | 0.50          | S/C+I/M |
| 22 <sup>nd</sup>    | ND(Lasota)      | 0.50          | D/W     |
| 24 <sup>th</sup>    | HPS             | 0.50          | I/M     |

# BREEDER VACCINATION SCHEDULE (Continued)

| Age(day)              | Vaccine         | Dose/Bird(ml)    | Route       |
|-----------------------|-----------------|------------------|-------------|
| 26 <sup>th</sup>      | IB              | 0.50             | D/W         |
| 38 <sup>th</sup>      | AI              | 0.50             | S/C+I/M     |
| 40 <sup>th</sup>      | I.B.D           | 0.50             | D/W         |
| 6 <sup>th</sup> week  | M.gallisepticum | 0.50             | S/C+I/M     |
| 7 <sup>th</sup> week  | ND(Lasota)      | 0.50             | D/W         |
| 8 <sup>th</sup> week  | Fowl pox        | One dip of prong | W/W         |
| 9 <sup>th</sup> week  | IB              | One dip of prong | D/W         |
| 10 <sup>th</sup> week | ILT             | 0.025            | S/C in neck |

## BREEDER VACCINATION SCHEDULE (Continued)

| Age(day)                         | Vaccine                    | Dose/Bird (ml) | Route         |
|----------------------------------|----------------------------|----------------|---------------|
| 11 <sup>th</sup> week            | Coryza                     | 0.50           | E/D           |
| 12 <sup>th</sup> week            | ND(Lasota)                 | 0.50           | D/W           |
| 13 <sup>th</sup> week            | Avian<br>Encephalomyelitis | 0.50           | D/W           |
| 14 <sup>th</sup> week            | IB                         | 0.50           | D/W           |
| 15 <sup>th</sup> week            | M.gallisepticum            | 0.50           | S/C +I/M      |
| 17 <sup>th</sup> week            | Coryza                     | 0.25           | S/C           |
| 21 <sup>st</sup> week            | ND+IB+IBD+EDS-oil<br>base  | 0.5            | I/M in breast |
| Repeat after every<br>two months | ND(Lasota)                 | 0.5            | D/W           |

# General Precaution for Vaccination

- ❑ Strictly follow the direction of the manufacturer
- ❑ Reliable source of vaccine
- ❑ Expiry date
- ❑ Transportation
- ❑ Storage of vaccines
- ❑ Quality of drinking water
- ❑ Time of vaccination

# Vaccine Handling

- ▶ Vaccines are biological in nature therefore very sensitive to environment.
- ▶ Damaging the vaccine components reduces the chance of stimulating the effective immune response.
- ▶ Improper vaccine handling=vaccine failure.
- ▶ Store at the temperature of 4c.
- ▶ Keep in dark covered container.
- ▶ Protect from UV light.



# Who should not be vaccinated ?

- ⊕ Allergy
- ⊕ Fever
- ⊕ HIV infection
- ⊕ Immunodeficiency
- ⊕ IG administration
- ⊕ Neurological disorder
- ⊕ Prematurity
- ⊕ Reactions to Previous vaccine
- ⊕ Simultaneous administration of Vaccines
- ⊕ Thrombocytopenia

# Vaccine Failure Causes



- ▶ USE OF EXPIRED VACCINE
- ▶ IMPROPER STORAGE OF VACCINES
- ▶ IMPROPER ROUTE OF VACCINATION
- ▶ GENETIC RESISTENCE
- ▶ HEALTH STATUS OF THE ANIMAL AND FLOCK
- ▶ IMMUNO SUPPRESSION DUE TO DRUGS
- ▶ MYCOTOXINS IN FEED
- ▶ WATER DEPRIVATION AND HEAT STRESS
- ▶ CONCURRENT DISEASE PROCESS
- ▶ PROLONG INTERVAL BETWEEN VACCINATION



# Vaccine Failure Causes



- ▶ ADMINISTRATION ERRORS QUALITY OF WATER
- ▶ MATERNAL ANTIBODIES
- ▶ PRESENCE OF VARIANT IN FIELD
- ▶ COLD AND HIGH DENSITY STRESS
- ▶ POOR NUTRITION
- ▶ PRESENCE OF AMMONIA IN HOUSES
- ▶ POOR ANTIGENICITY OF VACCINES
- ▶ VACCINE INTERFERENCE
- ▶ VACCINATION REACTIONS

# Conclusion

- ❑ Vaccines are one of the most effective health interventions ever developed.
- ❑ Three types of vaccines are currently used in humans: attenuated microorganisms, inactivated (killed) microorganisms, or purified macromolecules. Recombinant vector vaccine and Plasmid DNA vaccines are also used.
- ❑ They induce both humoral and cell-mediated immunity.
- ❑ Some boosters (called adjuvants) are also used in association with vaccines for increasing the immune response.
- ❑ As the vaccines have a lot of benefits, they do carry some harmful effects too.

# References

- ▶ <http://paksas.blogspot.com/2012/08/vaccination-schedule-for-poultry-birds.html#.WfR1Q2iCzIU>
- ▶ <https://content.pk/pakistan/vaccination-schedule-for-broiler-birds-layers-farms-in-pakistan/>
- ▶ <http://www.pakissan.com/english/allabout/livestock/poultry/causes.and.effects.of.vaccination.failure.shtml>
- ▶ <https://www.historyofvaccines.org/timeline>
- ▶ <https://www.vaccines.gov/basics/prevention/immunity/index.html>
- ▶ <https://www.historyofvaccines.org/content/articles/different-types-vaccines>
- ▶ <https://www.vaccines.gov/basics/types/index.html>